Scalar + Vector

A scalar is a quantity that has only **magnitude** (value) and NO **direction**.

Examples:

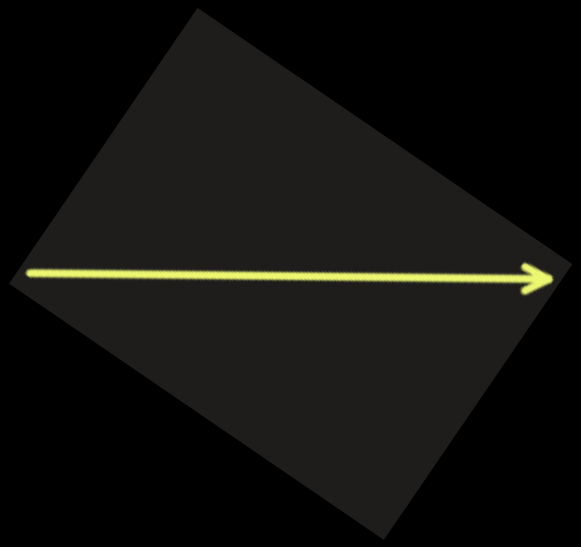
- 10 kilograms
- 25°C temperature
- 5 seconds
- ₹ 1000
- $342

All have size/value only — no direction.

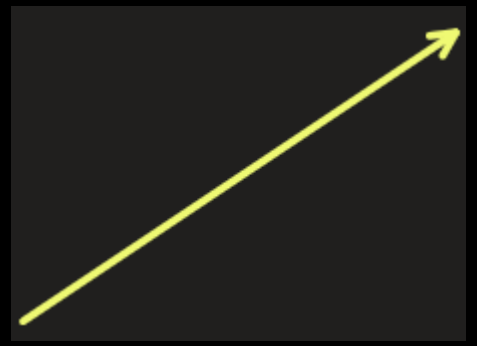
A **vector** is a quantity that has **both**:

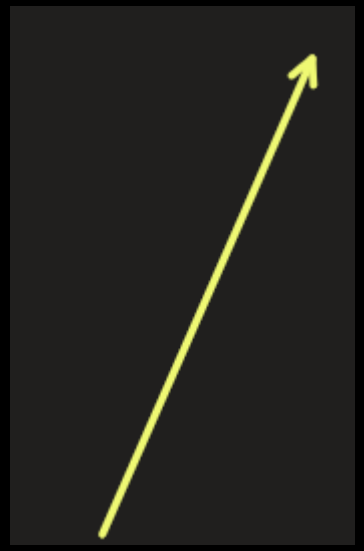
- **Magnitude** → how much (size or length)
- **Direction** → where it is going

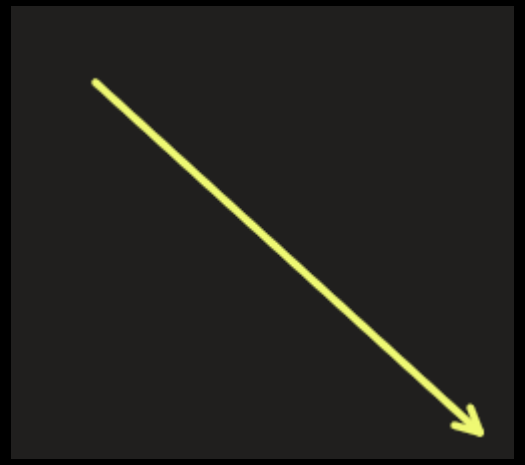
If a car moves:
60 km/h → magnitude
East → direction
This is a **vector** because direction matters.



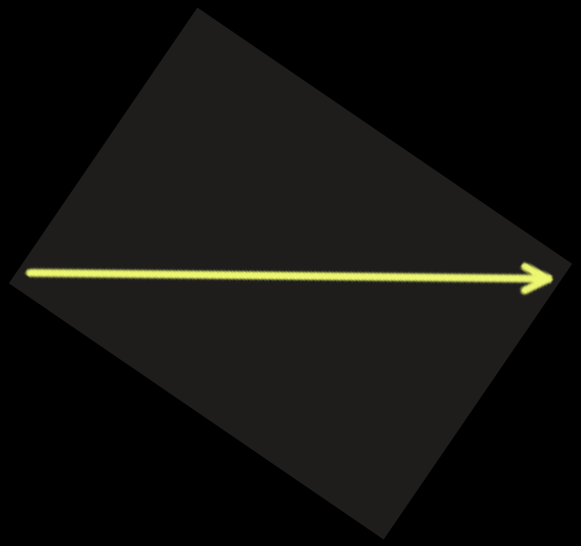
Example: 3 vectors shown below







Basic Idea
Scalar: 10
Vector: 10 meters **east**

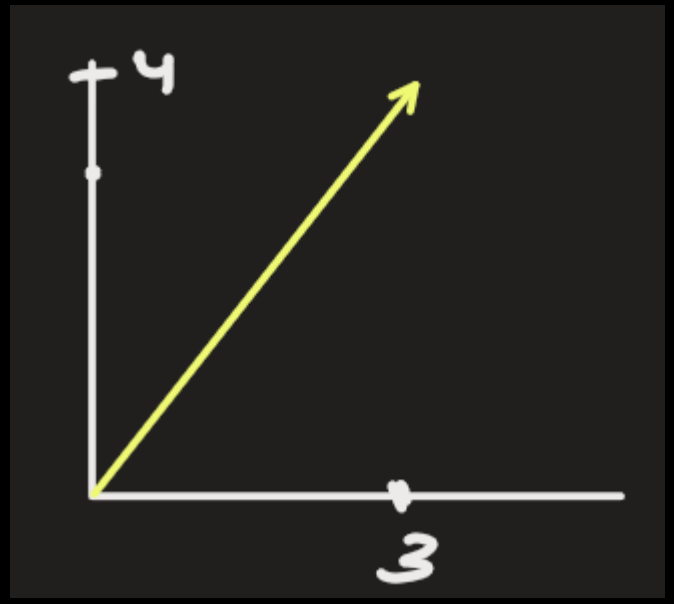


The direction makes it a vector.

Mathematically, a vector is often written as:

$$\vec{v} = (\mathrm{x}, \mathrm{y})$$

Example: (3,4)
Means move **3 units in x-direction** and then move **4 units in y-direction.**



In **Machine Learning**, a **vector** is simply:
A list (ordered collection) of numbers that represents data.

$$\vec{a} = (2, 3, 5)$$
$$\vec{b} = (3, 2, 5)$$

Here vector $\vec{a}$ and $\vec{b}$ are different.

Vectors are used to represent **features** of a data point.
Example:

- 2D vector: day data (temp=30C, humidity=40%) -> [30, 40]
- 3D vector: Patient data(height=170cm, weight=65Kg, age=30 years) → [170, 65, 30]
- 3D vector: student data (age=17, study-hours=3, percentage=78%) -> (17, 3, 78)
- 5D vector: (glucose, BMI, BP, age, skin) –> [13, 2, 12, 45, 0.3]
- **Word embeddings** in NLP (e.g., "king" → [0.21, -0.34, ..., 0.89])

3D vector: student data (age=17, study-hours=3, percentage=78%) -> (17, 3, 78)

This vector describes **one student**.
Each value = a **feature**.

Here features are age, study-hours, percentage.

So: **Vector = Feature Representation**

Example: Text → Vector (Very Important in ML)

Machines cannot read words like humans.

Sentence: I understand data science.

Converted into vector: [0.2, 0.8, 0.6, 0.1, ...]

This process is called:

- Feature encoding
- Embedding
- Vectorization

Example: Dataset = Collection of Vectors

Dataset looks like:

Age	Salary	Experience
22	30000	1
35	70000	10
52	56000	9

Vector representation is
[22, 30000, 1]
[35, 70000, 10]
[52, 56000, 9]

Each row = feature vector.

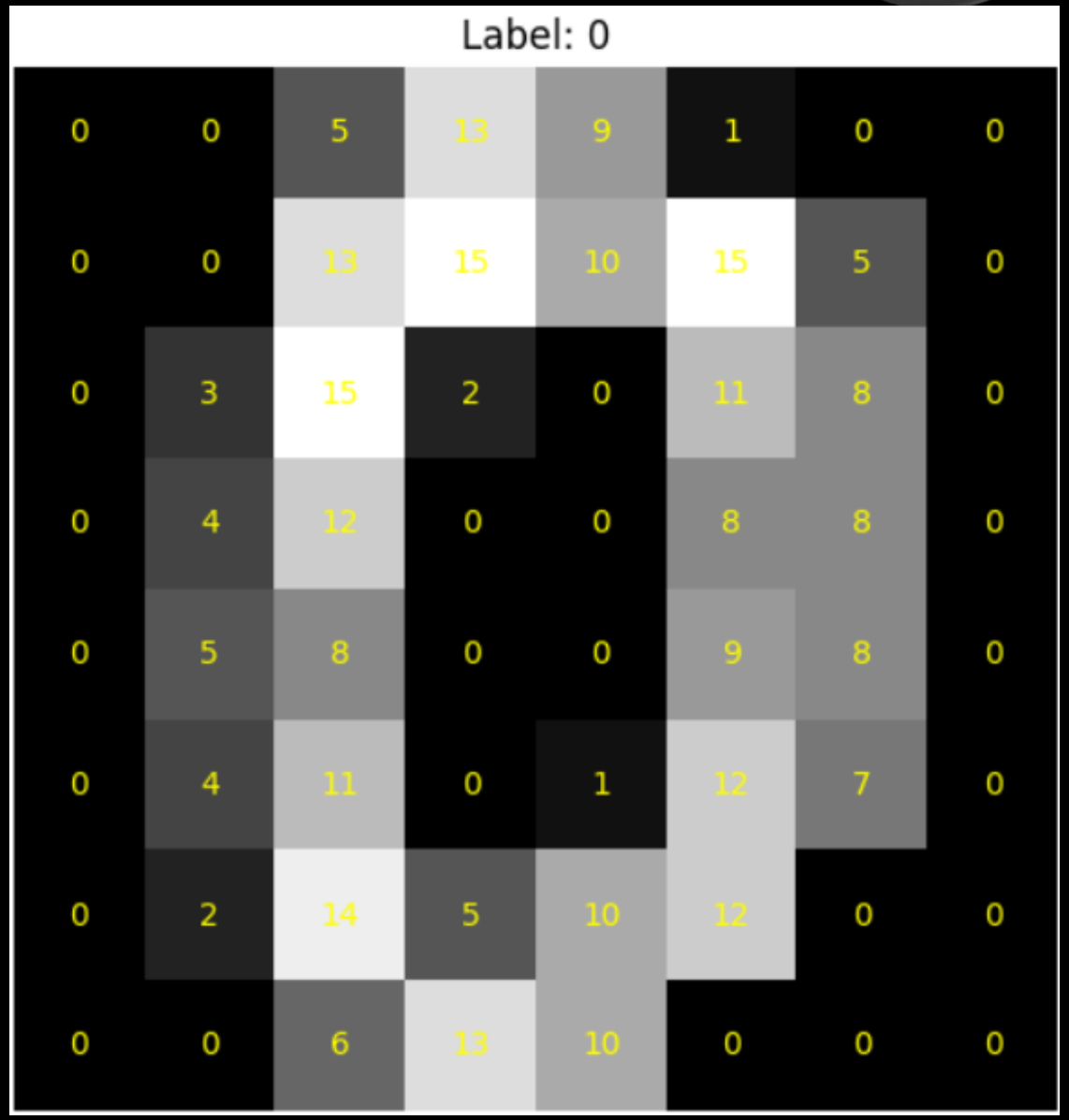
Example: Image as a Vector

An image is just pixels.

- Size of each image is 8x8 pixels
- Each image is represented by 8x8= 64 pixels
- The values of pixel range from 0 – 16.

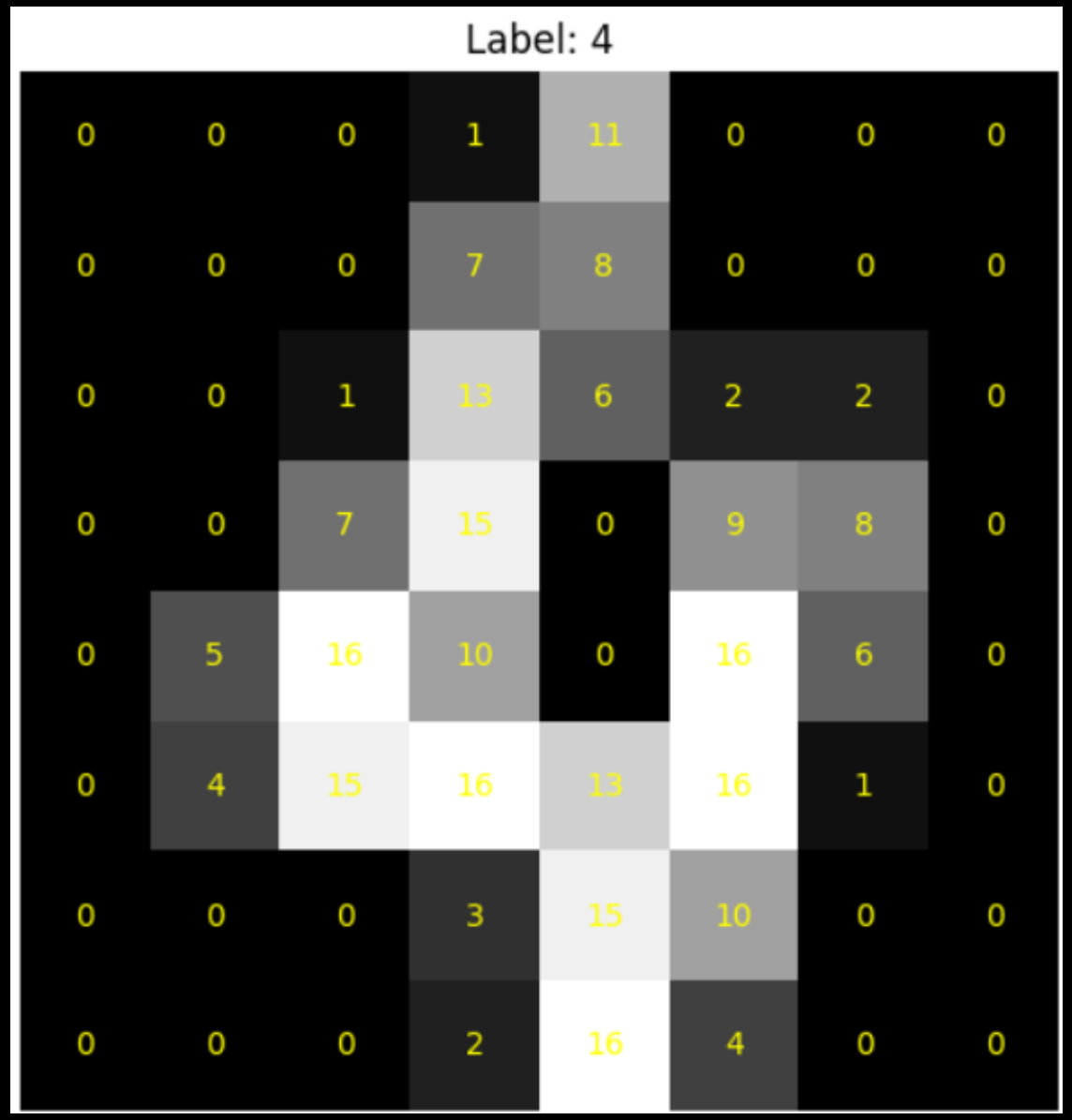
Digit 0 would contain following pixel values:

[0, 0, 5, 13, 9, 1, 0, 0, 0, 0, 13, 15, 10, 15, 5, 0,
0, 3, 15, 2, 0, 11, 8, 0, 0, 4, 12, 0, 0, 8, 8, 0, 0,
5, 8, 0, 0, 9, 8, 0, 0, 4, 11, 0, 1, 12, 7, 0, 0, 2,
14, 5, 10, 12, 0, 0, 0, 0, 6, 13, 10, 0, 0, 0,]

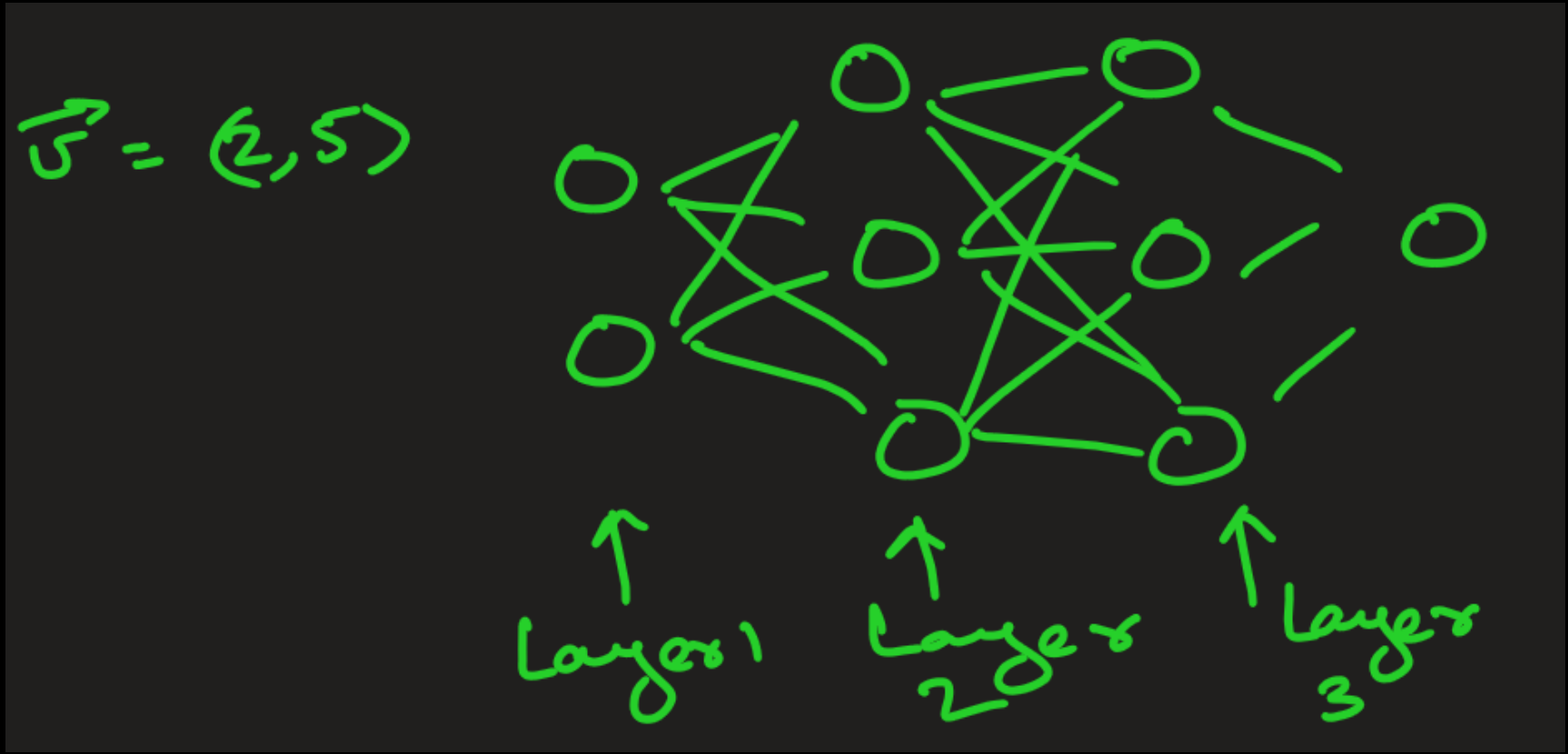


Digit 4 would contain following pixel values:

[0, 0, 0, 1, 11, 0, 0, 0, 0, 0, 0, 7, 8, 0, 0, 0,
0, 0, 1, 13, 6, 2, 2, 0, 0, 0, 7, 15, 0, 9, 8, 0,
0, 5, 16, 10, 0, 16, 6, 0, 0, 4, 15, 16, 13, 16, 1,
0, 0, 0, 0, 3, 15, 10, 0, 0, 0, 0, 0, 2, 16, 4, 0,
0,]

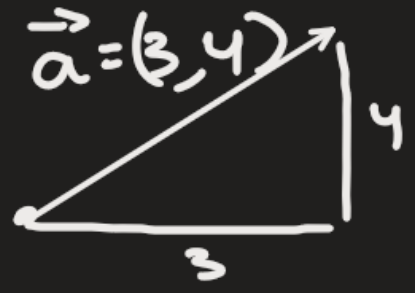


In **neural networks**, each layer transforms an input vector into another vector using weights and activation functions.



Magnitude of vector:

In 2D ->



$$|\vec{a}| = a$$
$$= \sqrt{3^2 + 4^2}$$
$$= \sqrt{25} = 5$$

This idea can be extended to higher dimension n-D

$$\vec{a} = (2, 3, 5, 7)$$
$$|\vec{a}| = a = \sqrt{2^2 + 3^2 + 5^2 + 7^2}$$

Scalar

A number that scales up or down the vector or any measurable quantity

For example:

2 * [170, 65, 30] = [340, 130, 60] …..this scales up by factor of 2

0.5 * [170, 65, 30] = [85, 32.5, 15] ….. This scales down the values by half

Why Vectors Are Important in ML

Machine learning works by comparing vectors.

Models compute:

- Distance between vectors
- Angle between vectors
- Similarity between vectors

Example:

- Recommendation systems
- Face recognition
- ChatGPT embeddings
- Search engines

Distance between 2 vectors:

$\vec{a} = (1, 3, 5) \rightarrow$ data points

$\vec{b} = (1, 5, 2) \rightarrow$

$$d(\vec{a}, \vec{b}) = \sqrt{(1-1)^2 + (3-5)^2 + (5-2)^2}$$

- Explain vectors and matrix are general form of numbers. Then explain dot product of vectors, matrix addition/subtraction, scalar multiplication (c*M),

Explain how 2 matrix gets multiplied.

Vector Space

- Matrices satisfy following few important properties which have similar counterpart in Q :

1) There is an element of V, denoted by 0, such that

$u + O = O + u = u$

for all elements u of V.

2) Given an element u of V, there exists an element -u in V such that

$u+(-u)=O$.

3) For all elements u of V, there exists an element 1 in V such that

$1 \cdot u = u$

4) There exist u_inv of u s.t.

$u * u_inv = 1$